CHEM1043 Tutorial 3

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Question 1

A beaker has a radius of 5 cm. (4 marks)

(a) Calculate the mass (in kg) if the beaker exerts 8.5 Torr on the surface.

(b) The beaker is then filled with hexane (d = 0.655 g/cm3). Calculate the height (cm) of hexane if it exerts the same pressure as filled with 15 cm of dichloromethane (d = 1.33 g/cm3).

$$(a)P = \frac{F}{A} = \frac{m \times g}{A}$$

$$8.5Torr \times \frac{101325pa}{760\,Torr} = \frac{m \times 9.807ms^{-1}}{\pi r^2}$$

$$11332.40pa = \frac{m \times 9.807ms^{-1}}{\pi 0.05^2}$$

m=0.908kg

$$(b)P = h \times d \times g$$

$$P_{hexane} = P_{dichloromethane}$$

$$h \times 0.655gcm^{-3} \times g = 15cm \times 1.33gcm^{-3} \times g$$

h=30.5cm

Question 2

A sulfur (S) sample (2,460 g) is burned and converted into SO_2 (g) (see the chemical reaction bellow). What is the volume (L) of SO_2 (g) collected at 50 °C and 910 mmHg? (3 marks)

$$S_{(s)} + O_{2(g)} = SO_{2(g)}$$

No. of moles of SO_2 = No. of moles of $S = \frac{2460g}{32.06gmol^{-1}} = 76.7mol$

PV=nRT

$$910mmHg \times \frac{1atm}{760mmHg} \times V = 76.7 \times 0.08206 \mathrm{dm^3}\ atm\ K^{-1}\ \mathrm{mol^{-1}} \times 323.15K$$

$$V = 1699.3L$$

Question 3

A 4.5 L container is filled with N2 (g) at 600 mmHg and 30 °C. 0.765 g of C_6H_{12} (g) is then injected into the container. (4 marks)

(a) What is the total pressure in the container?

(b) What are the partial pressures of N2 and of C_6H_{12}, respectively?

$$(a)\ n_{C6H16} = \frac{0.765g}{12.01 \times 6 + 16 \times 1} = 0.00908mol$$

$$P_{C6H16} = \frac{nRT}{V} = \frac{0.00908\text{mol} \times 0.8206\text{dm}^3\ atm\ K^{-1}\ \text{mol}^{-1} \times (30 + 273.15)K}{4.5L} = 38.1mmHg$$

$$P_{Total} = P_{N2} + P_{C6H16} = 600mmHg + 38.1mmHg = 638.1mmHg$$

$$(b)\ P_{N2} = 600mmHg$$

$$P_{C6H16} = 38.1mmHg$$

Question 4

$$2Na_2O_2\ (s) + 2CO_2\ (g) \rightarrow 2Na_2CO_3\ (s) + O_2\ (g)$$

This reaction is used in submarines and space vehicles to remove expired CO_2 (g) and to generate O_2 (g) for breathing. The rate of CO_2 consumed is 0.35 L/min at 25 °C and 800 mmHg.

If the CO_2 (g) and O_2 (g) in the above reaction are measured at the same temperature and pressure: (6 marks)

(a) What volume (mL) of O_2 (g) is produced per minute? Show your calculations.

(b) What is the rate of Na_2O_2 (s) consumed in grams per hour? Show your calculations.

Question 4

(a)

$No.\ of\ moles\ of\ CO_2\ consumed\ per\ minute$

$PV = nRT$

$$800mmHg \times \frac{101325pa}{760mmHg} \times 0.001m^3 \times 3.5\% = n \times 8.314\text{m}^3\ \text{pa}\ K^{-1}\ \text{mol}^{-1} \times 298.15K$$

$$n = 0.0151mol$$

$$n_{O2} = \frac{n_{CO2}}{2} = 0.00755mol$$

$PV = nRT$

$$800 \times \frac{101325}{760} \times V_{O2} = 0.00755 \times 8.314 \times (25 + 273.15)$$

$$V_{O2} = 175mL$$

Question 4

(b)

$No.\ of\ moles\ of\ CO_2\ consumed\ per\ minute = No.\ of\ moles\ of\ Na_2O_2\ consumed\ per\ minute$

Ratio of Na_2O_2 consumed per hour is

$0.0151 \times (22.99 \times 2 + 15.999 \times 2) \times 60 = 70.6 g/h$

Question 5

A balloon was filled with 40 mL of N2 (g) at an initial pressure (Pi). When the pressure is increased by 120 mmHg, the volume becomes 23.1 mL. (5 marks)
(a) Assuming constant temperature and no leakage, calculate Pi and the final pressure (Pf).
(b) The sealed balloon is taken to a high altitude and it inflates almost to bursting. Using the kinetic-molecular theory, explain how the density would change and the phenomenon using kinetic-molecular theory of gas.

(a)

$P_f = P_i + 120mmHg$
$P_iV_i = P_fV_f$
$P_i \times 40 = (P_i{+}120) \times 23.1$
$P_\mathrm{i} = 164\mathrm{mmHg}$

P_f=164+120=284mmHg

(b) Assume the number of moles of gas and temperature are kept constant. When atmospheric pressure decrease, air molecules will collide wall of balloon less frequently and with a smaller change in momentum. Since the gas molecules in the balloon would balance the atmospheric pressure. The volume of the balloon would increase to reduce the pressure inside until it balance the atmospheric pressure. Since the density equals to the mass/volume. With the mass unchanged and the volume increase, the density will decrease.

Question 6

A 5.2 L rigid cylinder contains H_2 (g) at 10.5 atm and 22 °C. Assuming ideal gas behavior and constant temperature, what mass of Ar (g) must be added to raise the total pressure to 90.2 atm? (3 marks)

$No.\ of\ moles\ of\ H_2\ consumed\ per\ minute$

$$n_{H2} = \frac{PV}{RT} = \frac{10.5\text{atm} \times 5.2L}{0.08206\text{dm}^3\ atm\ K^{-1}\ \text{mol}^{-1} \times (33 + 273.15)K} = 2.25\text{mol}$$

$$n_{total} = 2.25 \times \frac{90.2}{10.5} = 19.4mol$$

$No.\ of\ moles\ of\ Ne$ need to be added=19.4-2.25=17.15mol

$$M_{Ar} = 17.15 \times 39.95 = 685.14g$$

Question 7

Two ideal gases, A and B, are mixed in a 3 L container at 285 K. Before mixing, A occupies 18.8 L at 2.33 bar and 325 K; B occupies 3.72 L at 50.2 bar and 299 K. Assuming no reaction between A and B, calculate the final partial pressures (bar) of A and B, and the total pressure. Show your working. (5 marks)

$$\frac{P_i V_i}{T_i} = \frac{P_f V_f}{T_f}$$

$$P_{Af} = \frac{P_{Ai} V_{Ai} T_{Af}}{T_{Ai} V_{Af}} = \frac{2.33 \times 18.8 \times 285}{325 \times 3} = 12.8 bar$$

$$P_{Bf} = \frac{P_{Bi} V_{Bi} T_{Bf}}{T_{Bi} V_{Bf}} = \frac{50.2 \times 3.72 \times 285}{299 \times 3} = 59.3 bar$$

$$P_{TOTAL} = P_{Af} + P_{Bf} = 12.8 + 59.3 = 72.1 bar$$

Question 8

Using kinetic-molecular theory of gases, explain why a gas fills uniformly a container of any shape. (4 marks)

As a gas is made up of a large number of particles and they are in constant random motion with no intermolecular forces between gas particles, the particles are free to go anywhere. When they are allowed to expand, they cover the whole volume of container through their random motion. Since they all have average kinetic energy, they are uniformly distributed throughout the container.

Question 9

3.5 kg of dry ice (solid CO_2) is completely sublimed and collected over water at 21 °C and 1 atm. Calculate the partial pressure of CO_2 (in Torr) and the volume of CO_2 collected (in L). (4 marks)

$$P_{bar} = P_{CO2} + P_{H2O}$$

$$760Torr = P_{CO2} + 18.66Torr$$
$$P_{CO2} = 741.34Torr$$

$$No.\ of\ moles\ of\ CO_2 = \frac{3500}{44.01} = 79.5mol$$

$$PV = nRT$$

$$741.34 \times \frac{1}{760} \times V = 79.5 \times 0.08201 \times 294.15$$
$$V = 1966L$$

Vapour Pressure of Water at Various Temperatures

	Vapor Pressure	
t, °C	kPa	mmHg
19.0	2.198	16.49
21.0	2.448	18.66
23.0	2.811	21.08
25.0	3.170	23.78

Question 10

Calculate the average kinetic energy, $\overline{E_k}$, for $O_2(g)$ at 325 K and 1.00 atm. Show your calculations. (3 marks)

$$\bar{E}_k = \frac{3}{2} \times \left(\frac{RT}{N_A}\right) = \frac{3}{2} \times \left(\frac{8.31 \times 325}{6.022 \times 10^{-23}}\right) = 6.73 \times 10^{-21} \text{J}$$

Question 11

Consider a sample of H_2S (g) at 25℃ and 4.9 atm. Calculate the u_{rms} and use the equation bellow to calculate fraction (F) of molecules that have speed equal to this u_{rms}. (5 marks)

$$u_{rms} = \sqrt{\frac{3RT}{M}}$$

$$M_{H2S} = 1.008 \times 2 + 32.06 = 34.08\text{g/mol}$$

$$u_{rms} = \sqrt{\frac{3 \times 8.31 \times 298.15}{0.03408}} = 467.15m/s$$

$$F(u) = 4\pi\left(\frac{M}{2\pi RT}\right)^{\frac{3}{2}} \times u^2 \times e^{-\frac{M\text{u}^2}{2\pi RT}}$$

$$F(u) = 4\pi\left(\frac{0.03408}{2\pi \times 8.31 \times 298.15}\right)^{\frac{3}{2}} \times 462.3^2 \times e^{-\frac{0.03408 \times 462.3^2}{2 \times 8.31 \times 298.15}} = 1.98 \times 10^{-3}\text{s/m}$$

Question 12

Use both the van der Waals equation and ideal gas equation to calculate the pressure (in bar) exerted by 5 mol CH_4 (g) in a volume of 11.0 L at 298 K. For CH_4, a = 2.303 bar L^2 mol^{-2} and b = 0.0431 L mol^{-1}. (4 marks)

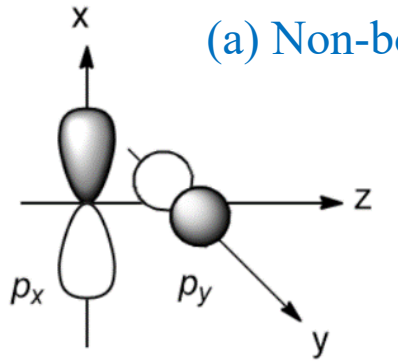
$$P_{\text{ideal}} = \frac{nRT}{V} = \frac{5 \times 0.08314 \times 298.15}{11} = 11.26\text{bar}$$

$$P_{van\ der\ Waals} = \frac{nRT}{V - \text{nb}} - \frac{n^2 \times a}{V^2} = \frac{5 \times 0.08314 \times 298.15}{11 - 5 \times 0.0431} - \frac{5^2 \times 2.303}{11^2} = 11.01\text{bar}$$

Question 13

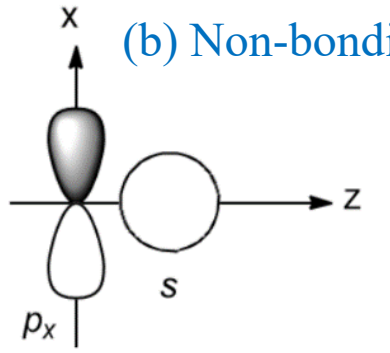
In each of the following combinations of atomic orbitals, identify whether the resulting molecular orbital (if any) is bonding, non-bonding or antibonding. For bonding/anti-bonding molecular orbitals, state whether it is σ or π as appropriate. (10 marks)

(a)



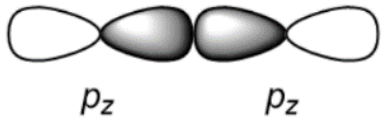
(a) Non-bonding

(b)



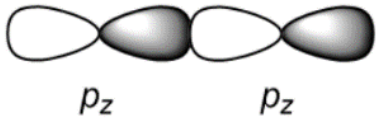
(b) Non-bonding

(c)



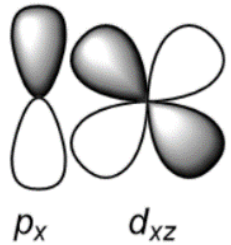
(c) σ bonding

(d)



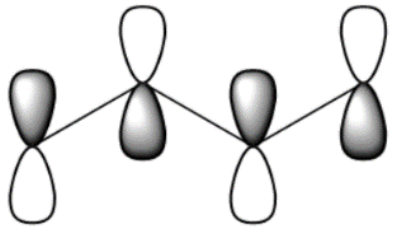
(d) σ antibonding

(e)



(e) π bonding

(f)



(a) π antibonding

Question 14

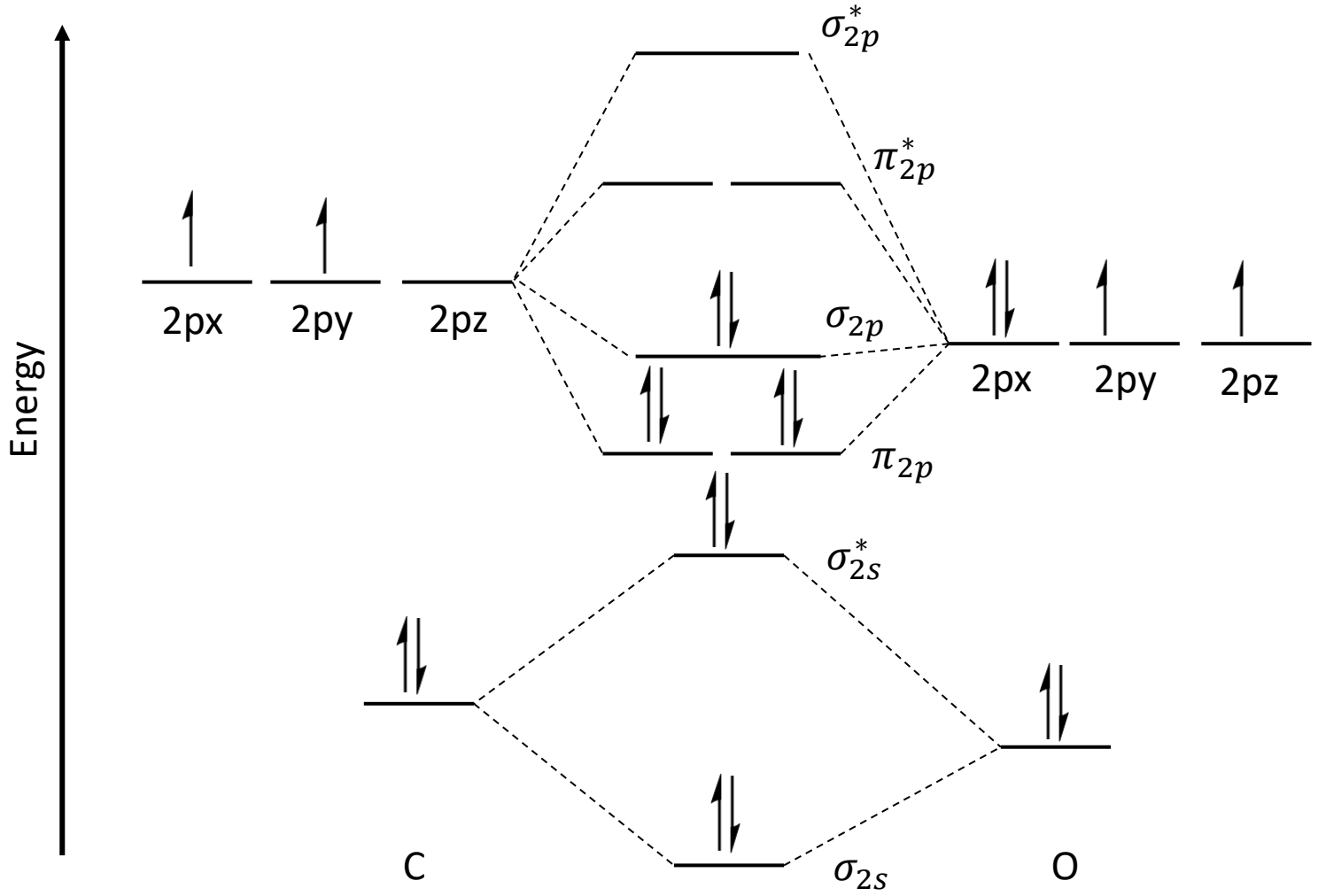
Consider CO, O_2^- and B_2. (14 marks)

(a) Represent the bonding with a fully labelled MO diagrams and atomic orbitals (AOs) in each case.

(b) Predict the bond order and magnetic property in each case.

(c) Compare the bond lengths and their relative stability (towards dissociation) of these three species. Explain briefly your answer.

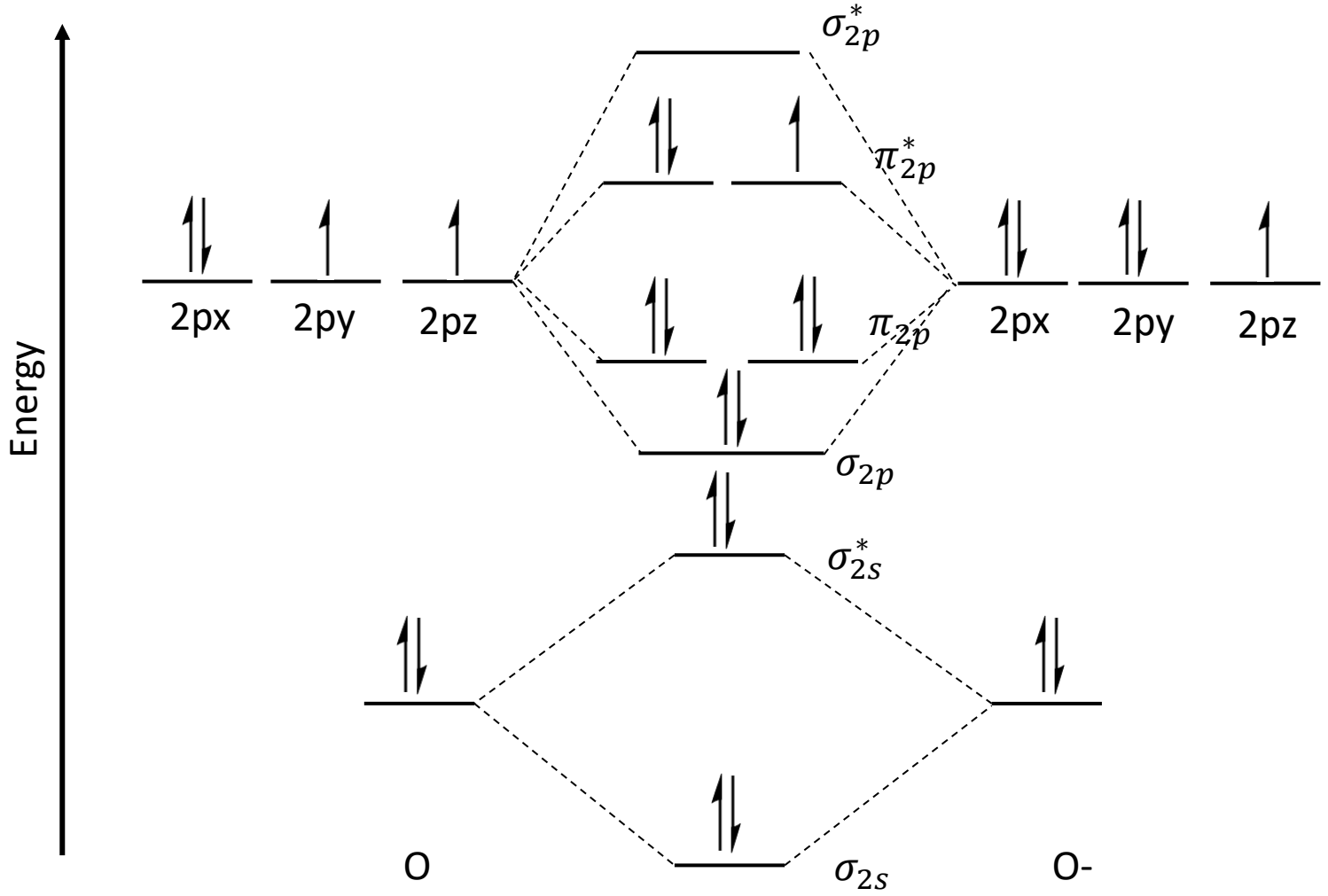
Question 14



Bond order = (8–2)/2 = 3

Diamagnetic

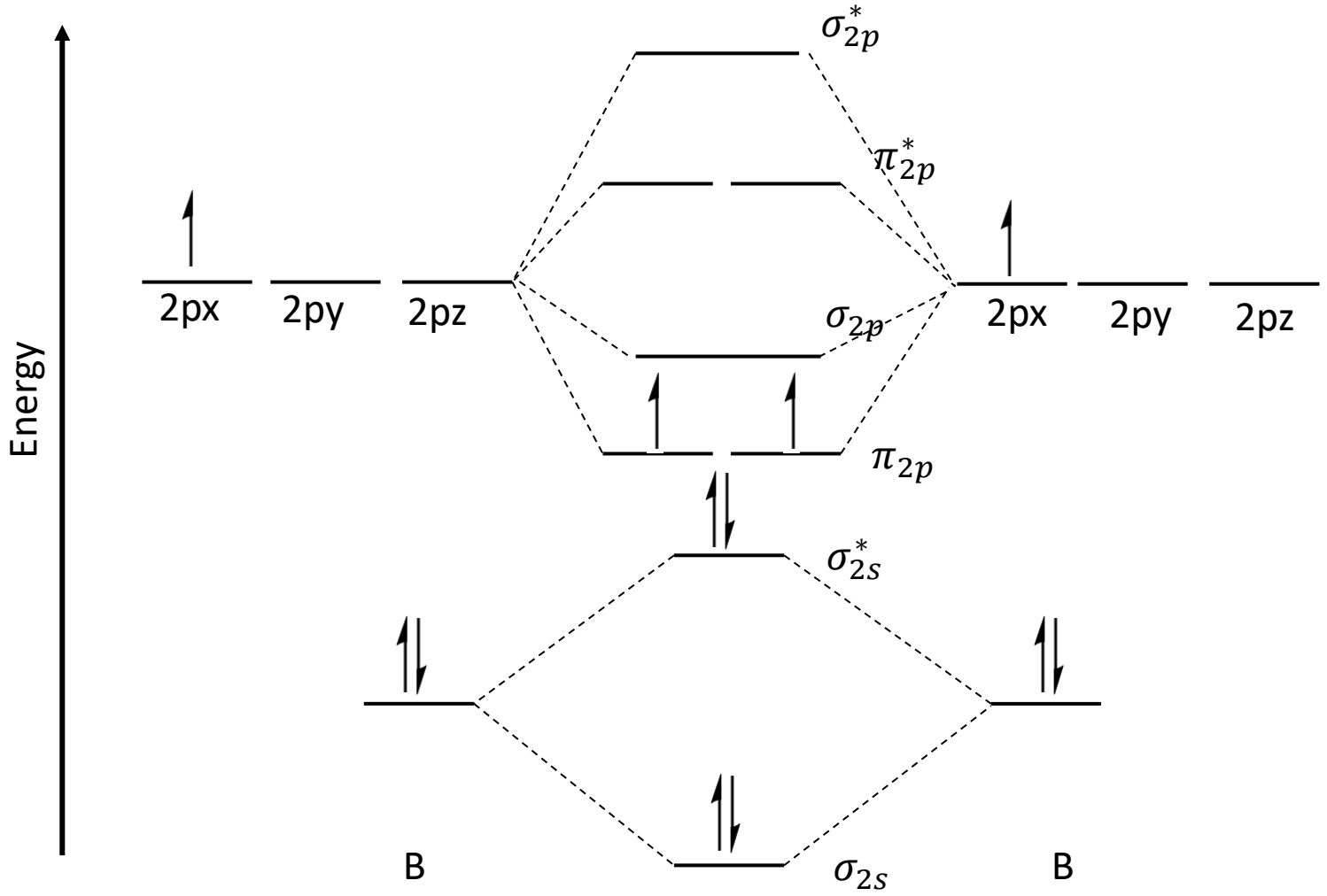
Question 14



Bond order = (8–5)/2 = 1.5

Paramagnetic

Question 14



Bond order = (4–2)/2 = 1

Paramagnetic

Question 14

(c) Compare the bond lengths and their relative stability (towards dissociation) of these three species. Explain briefly your answer.

(c) The larger bond order, the shorter bond length.

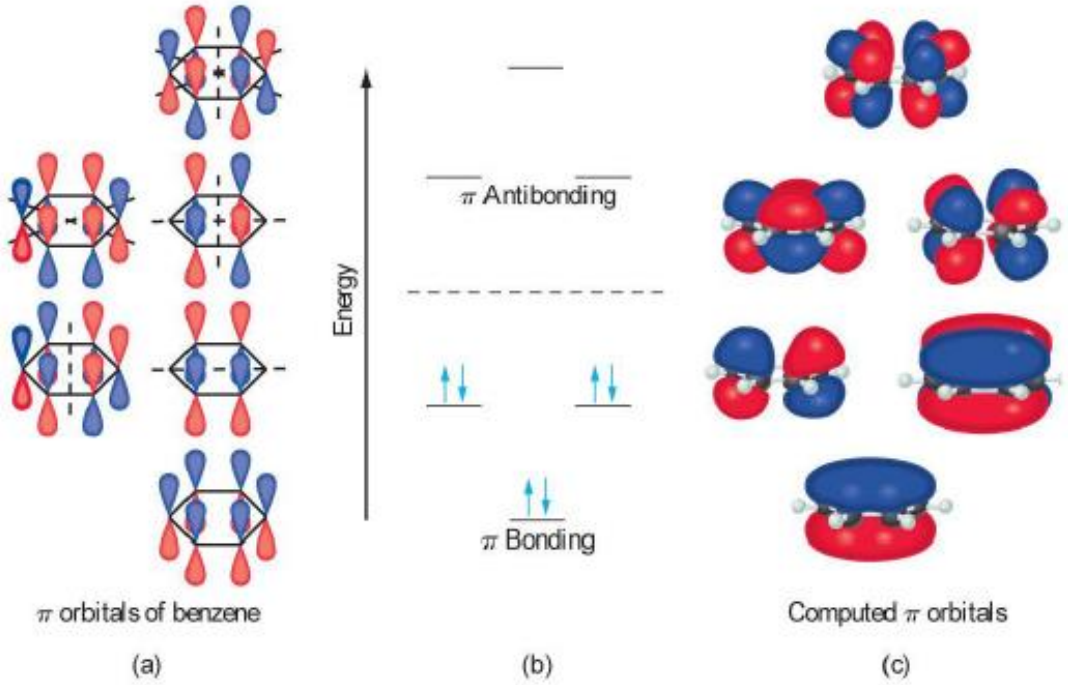
As the bond order is CO (3) > O2- (1.5) > B2 (1), the bond length is B2 > O2-> CO, predicted by MO theory.

The larger the bond order, the higher the stability.

Therefore, stability: CO (3) > O2- (1.5) > B2 (1).

Question 15

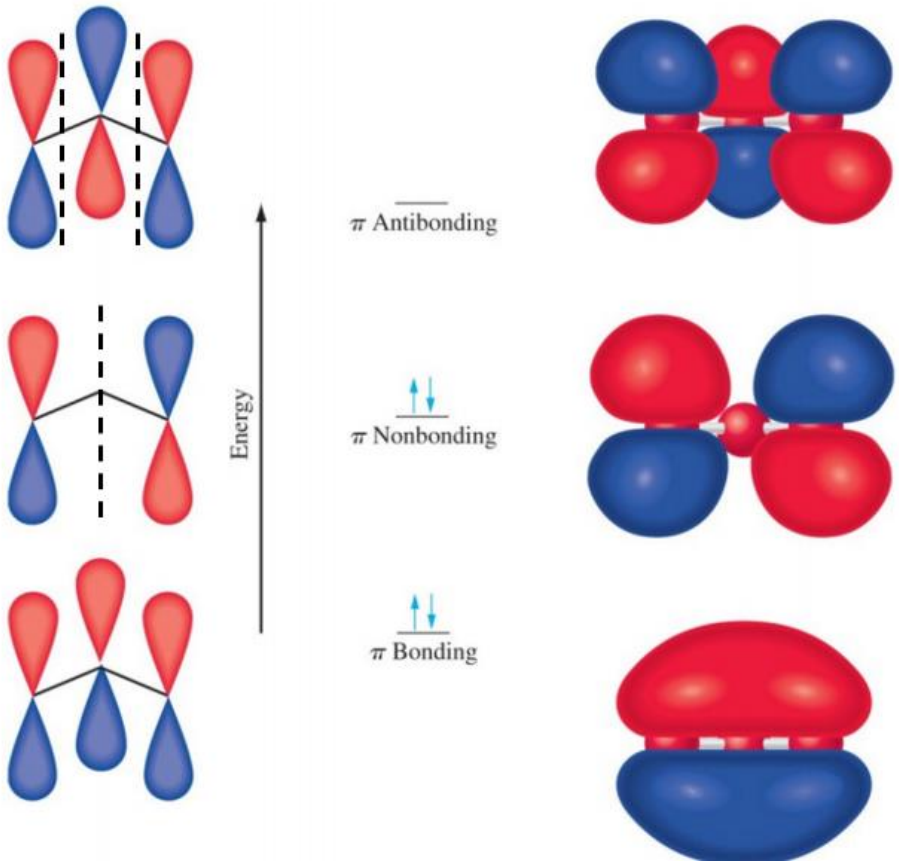
Draw the fully labeled π-system MO diagram and shapes of the MOs of benzene (4 marks). Indicate nodes in your drawings (1 marks). Calculate the total bond order for each C-C bond (1 marks).



The three bonding orbitals fill with 6 electrons, and the three antibonding orbitals remain empty. The bond order in this π system is (6–0)/2 = 3. The 3 bonds are distributed among the six C atoms, which amounts to 3/6 = 1/2 between each pair of C atoms. Together with the σ bonds, the total bond order for each C-C bond is 1.5.

Question 16

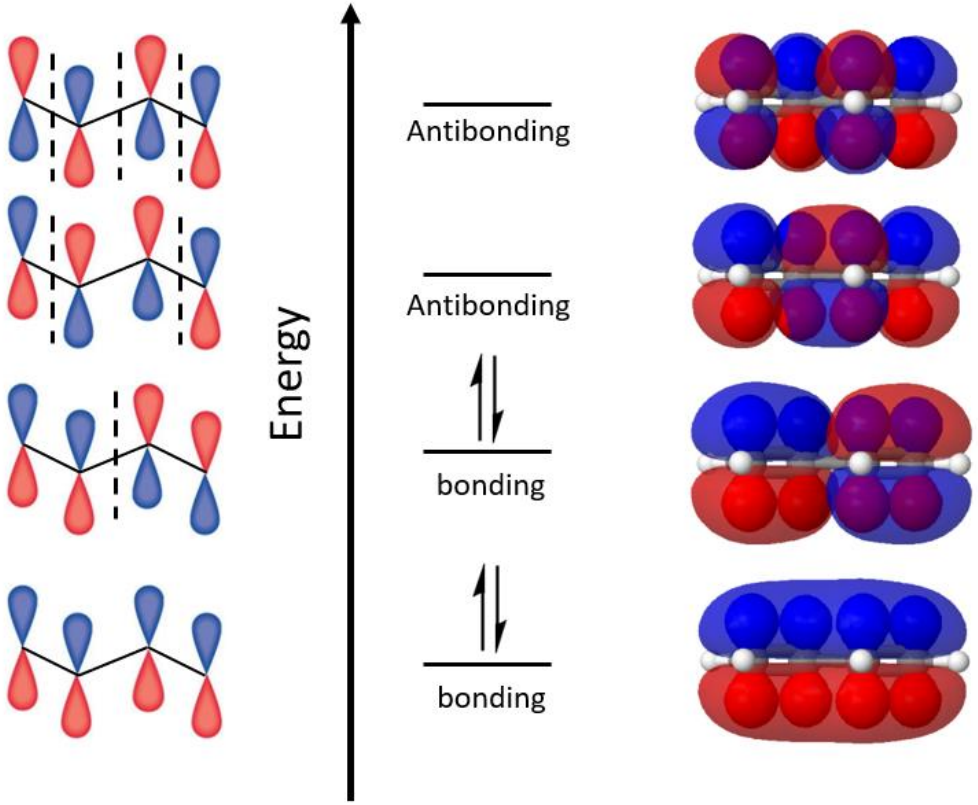
Draw the fully labeled π-system MO diagram and shapes of the MOs of ozone (4 marks). Indicate nodes in your drawings (1 marks). Calculate the total bond order for each C-C bond (1 marks).



The bond order associated with the π MOs is (2-0) / 2 = 1. This π bond is distributed between the two O−O bonds and amounts to one-half of a π bond for each. Together with the σ bonds, the total bond order for the O−O bond in O3 is 1.5.

Question 17

Draw the fully labeled π-system MO diagram and shapes of the MOs of trans 1,3-Butadiene (4 marks). Indicate nodes in your drawings (1 marks). Calculate the total π bond order of the π-system (1 marks).

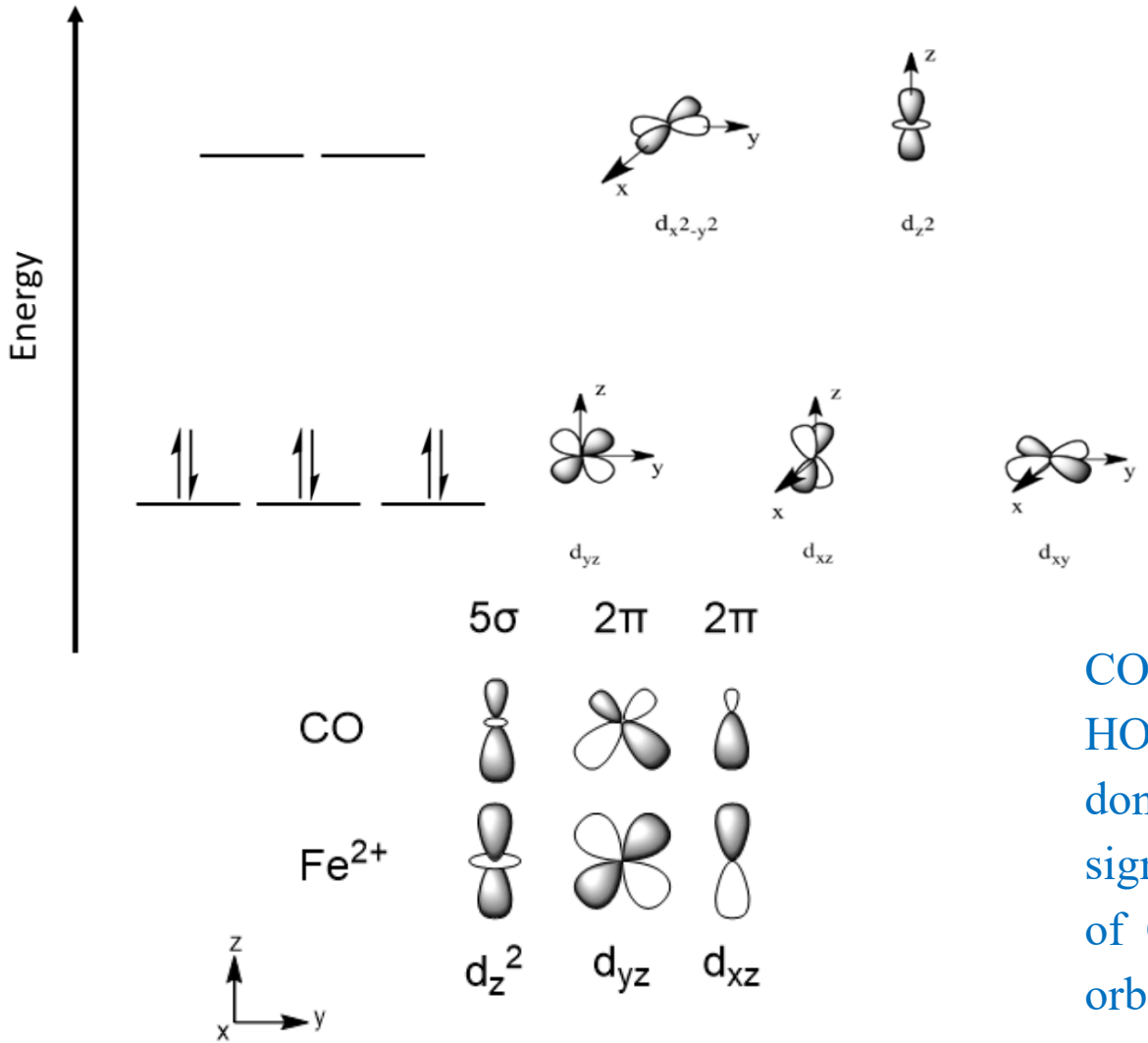


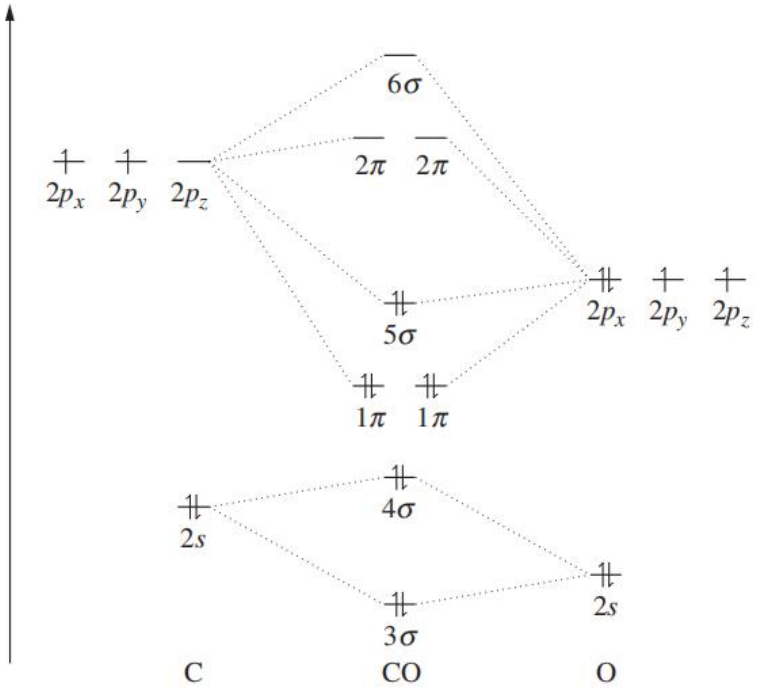
The bond order associated with the π MOs is (4-0) / 2 = 2

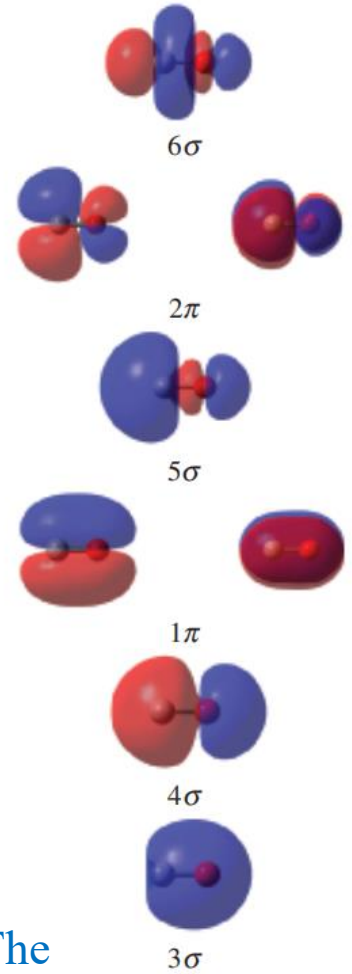
Question 18

Draw the five d-orbitals of Fe2+ in hemoglobin. The d_{z2} and d_{x2-y2} orbitals are degenerate and have higher energy than d_{xy}, d_{yz}, d_{xz} orbitals, which are also degenerate. Draw electron configurations in the d orbitals. Indicate which orbitals can form a sigma bond with carbon monoxide (CO) and can donate π electrons to CO, respectively. Indicate corresponding MOs of CO making the σ and π bond, respectively. (5 marks)

Question 18







CO molecule is a σ donor and π accepter. The HOMO of CO is σ bonding orbital which can donate electron to the dz2 of Fe2+ and form sigma bond. The LUMO π antibonding orbitals of CO will accept electron from the dxz, dyz orbitals in Fe2+ and form π bond

Question 19

Use band theory to explain the electrical conductivity of metals, semi-conductors and insulators, and explain how temperature affects the conductivity of semi-conductors. (7 marks)

For metals, the valence band is partially filled and hence it serves as conduction band while some have fully filled valence band but overlaps with conduction band. The ΔE between the occupied and unoccupied levels are so small that electrons can be excited from the highest filled state to unfilled levels easily, producing mobile electrons. Thus, metals conduct electricity.

For semi-conductors, the valence band is full and the conduction band is empty. The energy gap ΔE is small enough but thermal energy is needed for some electrons transitions. With temperature increases, electrons in valence band would acquire enough thermal energy to jump to a level in conduction band. The greater the thermal energy, the more electrons can make the transition. Hence, electrical conductivity increases with temperature.

The valence band of insulator is filled with electrons and the energy gap ΔE is large that it separates valence band from the conduction band. Few electrons can make transition between bands, making insulators do not conduct electricity.